Mohammad Nour

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Professional Summary

Third-year PhD researcher with 5 years of dedicated experience in hardware security and the application of machine learning to cryptographic analysis. Specialized in side-channel analysis with hands-on expertise in training deep learning models for key recovery, developing explainable-AI methods to interpret model behavior, and designing advanced metrics to maximize attack performance. Extensive practical experience with modern machine learning libraries and frameworks (e.g., PyTorch, TensorFlow, scikit-learn) for large-scale experimentation. Strong foundation in cryptography, information theory, and theoretical machine learning, combined with a proven ability to collaborate effectively within multidisciplinary research teams.

Skills

Programming Languages: Verilog, Python, C/C++, MATLAB, Shell scripting, solidity.

DL/ML: PyTorch, PyTorch Lightning; TensorFlow (v1/v2); OpenCV; scikit-learn; NumPy, Pandas.

Platforms & Tooling: Debian/GNU Linux, Windows; Git, Docker, bash; basic CUDA.

Projects

Uncertainty Estimation in Neural Network-enabled Side-channel Analysis

Conference on Cryptographic Hardware and Embedded Systems (CHES), 2026

- Developed a framework that quantifies predictive uncertainty in neural-network side-channel attacks, separating contributions from model behavior and data quality.

There's Waldo: PCB Tamper Forensic Analysis using Explainable AI on Impedance Signatures

IEEE International Symposium on Electromagnetic Compatibility, Signal and Power Integrity (EMC+SIPI), 2025

- Developed an explainable machine-learning framework that uses impedance (S-parameter) signatures to detect and classify PCB tampering with 96% accuracy.

Too hot to be true: Temperature calibration for higher confidence in NN-assisted side-channel analysis

Constructive Approaches for SeCurity Analysis and Design of Embedded systems (CASCADE), 2026

- Studied the effects of temperature calibration on deep learning based SCA.

Swarm in EM Hay: Particle Swarm-guided Probe Placement for EM SCA

USENIX WOOT Conference on Offensive Technologies, 2026

- Studied the effects of temperature calibration on deep learning based SCA.

Awards and Honers

- Recipient of the **DAC Young Fellow Award 2025.**
- Invited talk at **New England Hardware Security Day 2025**, MIT.
- **3rd Place Best Poster Award** at New England Hardware Security Day 2024, WPI.

Education

Worcester Polytechnic Institute — 2023 - Present

PhD - Electrical and Computer Engineering |Advisor: Prof. Fatemeh Ganji — *Worcester, MA*

Courses: *Applied cryptography & physical attacks, Physical Security of Microelectronic Systems*

Sharif University of Technology — 2020 - 2023

M.Sc. - secure communications and cryptography

Courses: *Deep learning, Cryptography, foundations of the blockchains*